Approximation study: paired differences between models

Differences are A - B for RMSE, maximum absolute error and fit time in seconds, per share and parameter budget. A negative RMSE or max-error difference means model A fits better; a negative seconds difference means A is faster. Pairing: seed s of A with seed s of B (deterministic models paired with every seed), so each cell is the mean and standard deviation of three paired differences. All values in the share's Min-Max price scale.

Caution on the seconds column. Fit times are not architecture-intrinsic: the KANs were fitted with 80 L-BFGS steps, the MLPs with 20,000 Adam steps (chosen because 4,000 was insufficient), splines by a direct linear solve. The seconds therefore mostly reflect those training-length choices and the single-CPU machine used, not a fair cost comparison. RMSE and max-error differences are the ones that bear on the paper.

KAN [1,1] - MLP of t

share	budget	params A	params B	d RMSE (mean +- sd)	d max|err| (mean +- sd)	d seconds (mean +- sd)
CBA.AX	15	15	13	-0.0046 +- 0.0005	-0.0262 +- 0.0079	-10.5 +- 0.5
CBA.AX	30	30	33	-0.0170 +- 0.0009	-0.0250 +- 0.0151	-10.0 +- 0.1
CBA.AX	60	60	61	-0.0294 +- 0.0074	-0.1078 +- 0.0120	-10.8 +- 0.6
CBA.AX	120	120	118	-0.0359 +- 0.0078	-0.1212 +- 0.0085	-9.9 +- 0.6
CBA.AX	240	240	253	-0.0287 +- 0.0024	-0.0743 +- 0.0164	-10.7 +- 2.6
TSLA	15	15	13	+0.0036 +- 0.0075	+0.0058 +- 0.0052	-11.1 +- 0.3
TSLA	30	30	33	-0.0163 +- 0.0010	-0.0570 +- 0.0053	-11.2 +- 0.2
TSLA	60	60	61	-0.0318 +- 0.0106	-0.1062 +- 0.0064	-9.8 +- 0.3
TSLA	120	120	118	-0.0299 +- 0.0028	-0.1231 +- 0.0302	-9.6 +- 0.8
TSLA	240	240	253	-0.0266 +- 0.0041	-0.0841 +- 0.0116	-9.6 +- 0.2
LLY	15	15	13	-0.0004 +- 0.0084	-0.0040 +- 0.0483	-10.0 +- 0.1
LLY	30	30	33	-0.0139 +- 0.0008	-0.0307 +- 0.0048	-9.5 +- 0.7
LLY	60	60	61	-0.0176 +- 0.0051	-0.0395 +- 0.0139	-10.1 +- 0.4
LLY	120	120	118	-0.0232 +- 0.0027	-0.0600 +- 0.0080	-9.7 +- 0.1
LLY	240	240	253	-0.0231 +- 0.0028	-0.0685 +- 0.0109	-9.5 +- 0.7

KAN [1,1] - Fourier MLP

share	budget	params A	params B	d RMSE (mean +- sd)	d max|err| (mean +- sd)	d seconds (mean +- sd)
CBA.AX	15	15	19	-0.0136 +- 0.0373	-0.1407 +- 0.2622	-9.3 +- 1.4
CBA.AX	30	30	31	-0.0115 +- 0.0189	-0.0486 +- 0.1061	-9.9 +- 0.1
CBA.AX	60	60	61	+0.0014 +- 0.0012	-0.0012 +- 0.0150	-11.1 +- 1.7
CBA.AX	120	120	127	-0.0005 +- 0.0016	-0.0026 +- 0.0069	-9.6 +- 0.5

share	budget	params A	params B	d RMSE (mean +- sd)	d max|err| (mean +- sd)	d seconds (mean +- sd)
CBA.AX	240	240	253	+0.0002 +- 0.0014	+0.0025 +- 0.0071	-10.6 +- 2.6
TSLA	15	15	19	-0.0081 +- 0.0398	-0.0677 +- 0.1913	-11.0 +- 0.2
TSLA	30	30	31	-0.0067 +- 0.0114	-0.0547 +- 0.0614	-11.6 +- 1.9
TSLA	60	60	61	+0.0011 +- 0.0022	+0.0149 +- 0.0155	-9.5 +- 0.1
TSLA	120	120	127	+0.0008 +- 0.0012	+0.0006 +- 0.0060	-10.0 +- 1.8
TSLA	240	240	253	-0.0005 +- 0.0013	+0.0045 +- 0.0160	-9.1 +- 0.4
LLY	15	15	19	-0.0185 +- 0.0309	-0.1288 +- 0.1751	-9.9 +- 0.2
LLY	30	30	31	-0.0170 +- 0.0175	-0.0860 +- 0.0908	-9.9 +- 1.5
LLY	60	60	61	-0.0011 +- 0.0005	-0.0083 +- 0.0042	-9.7 +- 0.2
LLY	120	120	127	-0.0005 +- 0.0015	-0.0025 +- 0.0025	-10.0 +- 1.2
LLY	240	240	253	-0.0010 +- 0.0007	-0.0081 +- 0.0048	-8.9 +- 0.7

KAN [1,1] - LSQ spline

share	budget	params A	params B	d RMSE (mean +- sd)	d max|err| (mean +- sd)	d seconds (mean +- sd)
CBA.AX	15	15	15	+0.0100 +- 0.0000	-0.0057 +- 0.0000	+1.3 +- 1.5
CBA.AX	30	30	30	+0.0051 +- 0.0000	+0.0080 +- 0.0000	+0.6 +- 0.1
CBA.AX	60	60	60	+0.0012 +- 0.0000	-0.0205 +- 0.0000	+0.9 +- 0.1
CBA.AX	120	120	120	-0.0002 +- 0.0000	-0.0020 +- 0.0000	+1.1 +- 0.2
CBA.AX	240	240	240	+0.0001 +- 0.0000	+0.0061 +- 0.0000	+2.6 +- 1.6
TSLA	15	15	15	+0.0244 +- 0.0000	+0.0610 +- 0.0000	+0.5 +- 0.0
TSLA	30	30	30	+0.0090 +- 0.0000	+0.0255 +- 0.0000	+0.6 +- 0.0
TSLA	60	60	60	+0.0019 +- 0.0000	+0.0223 +- 0.0000	+0.7 +- 0.1
TSLA	120	120	120	+0.0001 +- 0.0000	-0.0042 +- 0.0000	+1.2 +- 0.5
TSLA	240	240	240	+0.0004 +- 0.0000	+0.0010 +- 0.0000	+1.9 +- 0.0
LLY	15	15	15	+0.0122 +- 0.0000	+0.0286 +- 0.0000	+0.5 +- 0.0
LLY	30	30	30	+0.0045 +- 0.0000	+0.0051 +- 0.0000	+1.0 +- 0.6
LLY	60	60	60	+0.0006 +- 0.0000	-0.0011 +- 0.0000	+0.8 +- 0.1
LLY	120	120	120	+0.0006 +- 0.0000	+0.0002 +- 0.0000	+1.2 +- 0.1
LLY	240	240	240	+0.0002 +- 0.0000	-0.0012 +- 0.0000	+1.9 +- 0.6

KAN [1,h,1] - KAN [1,1]

share	budget	params A	params B	d RMSE (mean +- sd)	d max|err| (mean +- sd)	d seconds (mean +- sd)
CBA.AX	15	24	15	+0.0013 +- 0.0064	+0.0076 +- 0.0289	+0.1 +- 1.6
CBA.AX	30	24	30	+0.0256 +- 0.0064	+0.0473 +- 0.0289	+1.1 +- 0.3
CBA.AX	60	48	60	+0.0251 +- 0.0025	+0.0888 +- 0.0360	+4.6 +- 0.4
CBA.AX	120	120	120	+0.0274 +- 0.0041	+0.0982 +- 0.0173	+5.7 +- 0.2
CBA.AX	240	240	240	+0.0257 +- 0.0016	+0.0718 +- 0.0047	+7.1 +- 1.7
TSLA	15	24	15	-0.0028 +- 0.0061	-0.0172 +- 0.0387	+1.4 +- 0.6
TSLA	30	24	30	+0.0382 +- 0.0061	+0.0993 +- 0.0387	+1.7 +- 0.8
TSLA	60	48	60	+0.0297 +- 0.0057	+0.0854 +- 0.0154	+4.6 +- 1.2
TSLA	120	120	120	+0.0230 +- 0.0037	+0.0999 +- 0.0010	+6.0 +- 0.3
TSLA	240	240	240	+0.0269 +- 0.0044	+0.1122 +- 0.0071	+8.5 +- 0.2
LLY	15	24	15	-0.0010 +- 0.0036	+0.0033 +- 0.0178	+2.4 +- 1.2
LLY	30	24	30	+0.0220 +- 0.0036	+0.0682 +- 0.0178	+1.8 +- 1.7
LLY	60	48	60	+0.0142 +- 0.0043	+0.0365 +- 0.0170	+5.3 +- 0.5
LLY	120	120	120	+0.0114 +- 0.0018	+0.0326 +- 0.0084	+6.2 +- 0.9
LLY	240	240	240	+0.0160 +- 0.0017	+0.0432 +- 0.0073	+8.8 +- 1.0

Fourier MLP - MLP of t

share	budget	params A	params B	d RMSE (mean +- sd)	d max|err| (mean +- sd)	d seconds (mean +- sd)
CBA.AX	15	19	13	+0.0090 +- 0.0377	+0.1145 +- 0.2661	-1.2 +- 0.9
CBA.AX	30	31	33	-0.0055 +- 0.0181	+0.0236 +- 0.0981	-0.0 +- 0.1
CBA.AX	60	61	61	-0.0307 +- 0.0086	-0.1066 +- 0.0265	+0.3 +- 2.3
CBA.AX	120	127	118	-0.0354 +- 0.0088	-0.1187 +- 0.0142	-0.3 +- 0.3
CBA.AX	240	253	253	-0.0289 +- 0.0022	-0.0769 +- 0.0093	-0.1 +- 0.6
TSLA	15	19	13	+0.0117 +- 0.0472	+0.0734 +- 0.1905	-0.1 +- 0.5
TSLA	30	31	33	-0.0096 +- 0.0122	-0.0023 +- 0.0662	+0.4 +- 1.7
TSLA	60	61	61	-0.0329 +- 0.0094	-0.1212 +- 0.0106	-0.3 +- 0.2
TSLA	120	127	118	-0.0306 +- 0.0039	-0.1237 +- 0.0362	+0.3 +- 1.1
TSLA	240	253	253	-0.0261 +- 0.0048	-0.0886 +- 0.0185	-0.5 +- 0.2

share	budget	params A	params B	d RMSE (mean +- sd)	d max|err| (mean +- sd)	d seconds (mean +- sd)
LLY	15	19	13	+0.0182 +- 0.0390	+0.1248 +- 0.2210	-0.1 +- 0.1
LLY	30	31	33	+0.0031 +- 0.0182	+0.0552 +- 0.0955	+0.4 +- 1.1
LLY	60	61	61	-0.0165 +- 0.0047	-0.0312 +- 0.0124	-0.4 +- 0.2
LLY	120	127	118	-0.0227 +- 0.0028	-0.0575 +- 0.0055	+0.3 +- 1.4
LLY	240	253	253	-0.0221 +- 0.0025	-0.0604 +- 0.0090	-0.6 +- 0.1

Paired t-tests (one-sample t-test of the cell-mean differences against zero; 'all 15 cells' pools the three shares and five budgets, per-share tests use the five budgets). With n = 5 or 15 these are indicative only; cells at different budgets are not independent draws from one population, so treat p-values as descriptive.

comparison	metric	scope	n	mean difference	t	p
KAN [1,1] - MLP of t	RMSE	all 15 cells	15	-0.0197	-6.43	0.0000
KAN [1,1] - MLP of t	RMSE	CBA.AX	5	-0.0231	-4.17	0.0140
KAN [1,1] - MLP of t	RMSE	TSLA	5	-0.0202	-3.09	0.0365
KAN [1,1] - MLP of t	RMSE	LLY	5	-0.0156	-3.72	0.0205
KAN [1,1] - MLP of t	max |err|	all 15 cells	15	-0.0615	-5.74	0.0001
KAN [1,1] - MLP of t	max |err|	CBA.AX	5	-0.0709	-3.55	0.0239
KAN [1,1] - MLP of t	max |err|	TSLA	5	-0.0729	-3.23	0.0320
KAN [1,1] - MLP of t	max |err|	LLY	5	-0.0405	-3.56	0.0236
KAN [1,1] - MLP of t	seconds	all 15 cells	15	-10.1220	-68.30	0.0000
KAN [1,1] - MLP of t	seconds	CBA.AX	5	-10.3613	-54.36	0.0000
KAN [1,1] - MLP of t	seconds	TSLA	5	-10.2573	-29.14	0.0000
KAN [1,1] - MLP of t	seconds	LLY	5	-9.7473	-76.13	0.0000
KAN [1,1] - Fourier MLP	RMSE	all 15 cells	15	-0.0050	-2.79	0.0145
KAN [1,1] - Fourier MLP	RMSE	CBA.AX	5	-0.0048	-1.51	0.2063
KAN [1,1] - Fourier MLP	RMSE	TSLA	5	-0.0027	-1.38	0.2406
KAN [1,1] - Fourier MLP	RMSE	LLY	5	-0.0076	-1.83	0.1404
KAN [1,1] - Fourier MLP	max |err|	all 15 cells	15	-0.0351	-2.69	0.0174
KAN [1,1] - Fourier MLP	max |err|	CBA.AX	5	-0.0381	-1.40	0.2355
KAN [1,1] - Fourier MLP	max |err|	TSLA	5	-0.0205	-1.21	0.2930
KAN [1,1] - Fourier MLP	max |err|	LLY	5	-0.0467	-1.82	0.1430
KAN [1,1] - Fourier MLP	seconds	all 15 cells	15	-10.0053	-50.64	0.0000
KAN [1,1] - Fourier MLP	seconds	CBA.AX	5	-10.1187	-30.37	0.0000
KAN [1,1] - Fourier MLP	seconds	TSLA	5	-10.2233	-22.05	0.0000
KAN [1,1] - Fourier MLP	seconds	LLY	5	-9.6740	-47.50	0.0000
KAN [1,1] - LSQ spline	RMSE	all 15 cells	15	+0.0047	+2.65	0.0189
KAN [1,1] - LSQ spline	RMSE	CBA.AX	5	+0.0032	+1.67	0.1699

comparison	metric	scope	n	mean difference	t	p
KAN [1,1] - LSQ spline	RMSE	TSLA	5	+0.0072	+1.56	0.1947
KAN [1,1] - LSQ spline	RMSE	LLY	5	+0.0036	+1.57	0.1905
KAN [1,1] - LSQ spline	max |err|	all 15 cells	15	+0.0082	+1.64	0.1235
KAN [1,1] - LSQ spline	max |err|	CBA.AX	5	-0.0028	-0.55	0.6121
KAN [1,1] - LSQ spline	max |err|	TSLA	5	+0.0211	+1.83	0.1406
KAN [1,1] - LSQ spline	max |err|	LLY	5	+0.0063	+1.11	0.3280
KAN [1,1] - LSQ spline	seconds	all 15 cells	15	+1.1200	+7.08	0.0000
KAN [1,1] - LSQ spline	seconds	CBA.AX	5	+1.2900	+3.73	0.0202
KAN [1,1] - LSQ spline	seconds	TSLA	5	+0.9927	+3.69	0.0210
KAN [1,1] - LSQ spline	seconds	LLY	5	+1.0773	+4.42	0.0115
KAN [1,h,1] - KAN [1,1]	RMSE	all 15 cells	15	+0.0188	+6.04	0.0000
KAN [1,h,1] - KAN [1,1]	RMSE	CBA.AX	5	+0.0210	+4.25	0.0131
KAN [1,h,1] - KAN [1,1]	RMSE	TSLA	5	+0.0230	+3.33	0.0292
KAN [1,h,1] - KAN [1,1]	RMSE	LLY	5	+0.0125	+3.29	0.0302
KAN [1,h,1] - KAN [1,1]	max |err|	all 15 cells	15	+0.0585	+5.63	0.0001
KAN [1,h,1] - KAN [1,1]	max |err|	CBA.AX	5	+0.0627	+3.85	0.0182
KAN [1,h,1] - KAN [1,1]	max |err|	TSLA	5	+0.0759	+3.21	0.0326
KAN [1,h,1] - KAN [1,1]	max |err|	LLY	5	+0.0368	+3.53	0.0242
KAN [1,h,1] - KAN [1,1]	seconds	all 15 cells	15	+4.3578	+6.05	0.0000
KAN [1,h,1] - KAN [1,1]	seconds	CBA.AX	5	+3.7387	+2.79	0.0493
KAN [1,h,1] - KAN [1,1]	seconds	TSLA	5	+4.4407	+3.31	0.0297
KAN [1,h,1] - KAN [1,1]	seconds	LLY	5	+4.8940	+3.78	0.0194
Fourier MLP - MLP of t	RMSE	all 15 cells	15	-0.0146	-3.16	0.0069
Fourier MLP - MLP of t	RMSE	CBA.AX	5	-0.0183	-2.14	0.0992
Fourier MLP - MLP of t	RMSE	TSLA	5	-0.0175	-2.09	0.1044
Fourier MLP - MLP of t	RMSE	LLY	5	-0.0080	-1.00	0.3753
Fourier MLP - MLP of t	max |err|	all 15 cells	15	-0.0264	-1.18	0.2567
Fourier MLP - MLP of t	max |err|	CBA.AX	5	-0.0328	-0.74	0.5019

comparison	metric	scope	n	mean difference	t	p
Fourier MLP - MLP of t	max |err|	TSLA	5	-0.0525	-1.37	0.2434
Fourier MLP - MLP of t	max |err|	LLY	5	+0.0062	+0.17	0.8726
Fourier MLP - MLP of t	seconds	all 15 cells	15	-0.1167	-1.01	0.3303
Fourier MLP - MLP of t	seconds	CBA.AX	5	-0.2427	-0.97	0.3865
Fourier MLP - MLP of t	seconds	TSLA	5	-0.0340	-0.18	0.8629
Fourier MLP - MLP of t	seconds	LLY	5	-0.0733	-0.38	0.7251